Where Delhi's Winter Smog Comes From: An Eight-Season Study

Summary

Over eight burning seasons (2018–2025), crop-residue fires in Punjab and Haryana account for about **15% of Delhi's October–November PM2.5** (95% confidence interval 7–22%). That is roughly 26 of the 180 μg/m³ people breathe on an average day of the season. It is a real and substantial share, but most of the pollution has other sources.

Every autumn the same argument returns: is it the farm fires, is it Diwali, or is it the weather? This study tries to answer that with the data that already exists:

- 280,512 hourly readings from four government monitors in Delhi;
- 742,861 NASA satellite fire detections;
- eight years of daily weather.

The core idea is simple. On some days the wind blows from the fields towards Delhi; on other days the same fires burn while the wind carries their smoke elsewhere. Comparing those days within the same week shows how much the smoke adds.

What the study found, stated as carefully as the evidence allows:

- **Crop fires matter, and the evidence that they cause pollution is strong.** When fires are upwind, Delhi's PM2.5 rises. When the same fires are downwind, it does not. Shuffling the wind randomly reproduces a result this large only 2% of the time.
- **Falling fire counts did not mean less smoke.** Satellite detections dropped about 90% between 2021 and 2025, yet the smoke's share of Delhi's pollution stayed near 15%. One detection now stands for much more smoke than before. This fits reports that burning has shifted to times the satellite does not see, though the study cannot prove that directly.
- **Diwali is a severe but brief spike.** On Diwali night, PM2.5 is about 2.3 times an ordinary night, and this happened in all eight years. Because it lasts only hours, it adds roughly 10% to the week that follows and under 2% to the season. Dussehra left no measurable mark.
- **Weather decides how much stays.** Within a single week, a still, stagnant day carries about 53 μg/m³ more PM2.5 than a well-ventilated one.
- **Starting the emergency plan earlier would change little.** Enforcing GRAP from 1 September, a month before burning begins, would lower season PM2.5 by at most 3.4–3.7%, and probably by nothing. Real cuts in October would do more.

These are estimates with uncertainty, from four monitors in one city. The sections below explain how each number was reached, which tests it passed, and the two tests it did not pass cleanly.

1. Why this question matters

For two months every year, Delhi's air moves from bad to dangerous. Between 2018 and 2025, PM2.5 averaged about 123 µg/m³ in October and 238 µg/m³ in November, roughly two to four times India's 24-hour standard of 60 µg/m³. Schools close, flights are delayed, and doctors see more patients with breathing problems.

The public conversation usually settles on one culprit at a time. Some years it is the farm fires in Punjab and Haryana; some years it is Diwali crackers; often it is simply "the weather". Each explanation leads to a different response. If fires dominate, the answer lies in the fields. If festivals dominate, it lies in fireworks rules. If weather dominates, cities can only prepare and protect people.

A measured answer matters because policy follows the explanation. This study asks two questions:

1. **What contributes to Delhi's October–November PM2.5, and by how much?** It covers crop fires, Diwali, Dussehra and the weather.
2. **Would acting about a month earlier help, and by how much?**

The second question follows from earlier work. An evaluation of the Graded Response Action Plan (GRAP), Delhi's emergency pollution plan, found no statistically significant drop in PM2.5 when its strictest stages were in force (+11.7%, 95% CI −9.4% to +25.5%). If the emergency response does not measurably help once pollution is severe, it is fair to ask whether starting earlier would.

2. What earlier work tells us

The size of the farm-fire share has been argued over for years, partly because different methods answer slightly different questions.

Source	Method	Reported crop-fire share
SAFAR daily estimates, as reported by Down To Earth (2021)	Forecast model, day by day	0–58% on individual days; 25–48% during 4–13 November 2021
Down To Earth estimate over the season (2021)	Averaging daily shares over Oct–Nov	13–17% of the season's PM2.5 load

Source	Method	Reported crop-fire share
Centre for Science and Environment analysis (2022)	81 CPCB stations with SAFAR fire data	About 28 µg/m³ a day over 52 days; over 80% of daily PM2.5 from non-stubble sources
Government affidavit to the Supreme Court (2021), as reported above	Not disclosed	4% in winter
TERI–ARAI source apportionment (2018)	Receptor and dispersion modelling	Delhi's own emissions 36% of winter PM2.5; the rest from NCR and beyond

Two lessons come out of this table. Daily shares and seasonal shares are very different numbers: a single smoky day can be half smoke while the season as a whole is not. And most published figures come from forecast models, whose assumptions are hard to check from outside.

Two more pieces of earlier work matter here:

- **Fire counts and satellite timing.** A study by ISRO's Space Applications Centre used geostationary satellites that observe every 15 minutes. It found that peak burning in Punjab and Haryana shifted from about 13:30 in 2020 to about 17:00 in 2024, after the overpass of the polar-orbiting satellites that most fire counts rely on (The Tribune). If true, falling satellite fire counts would overstate the real fall in burning.
- **Diwali chemistry.** Researchers at IIT Delhi measured 43 elements in PM2.5 around Diwali 2019. They found that fireworks made up about 92–95% of the metal content during the festival, and that the haze in the days after was driven mainly by biomass burning (Manchanda et al.).

This study adds an estimate that does not depend on a forecast model. It uses the observed day-to-day link between where the wind blows from and what Delhi's monitors record, over eight seasons.

3. The data, and the problem found in it

Everything in this study comes from public sources.

Data	Source	Coverage	Size
Air quality, hourly	CPCB Continuous Ambient Air Quality Monitoring	Anand Vihar, Ashok Vihar, Aya Nagar and Bawana, 2018–2025	280,512 hourly rows from 31 files

Data	Source	Coverage	Size
Air quality, 15-minute	CPCB / DPCC	Anand Vihar, 2024–2025	70,177 rows
Fire detections	NASA FIRMS, VIIRS S-NPP satellite	Punjab–Haryana, 2018–2025	742,861 vegetation-fire detections
Fire detections, second satellite	NASA FIRMS, VIIRS NOAA-20	Same area, 2024–2025	115,271 detections
Weather	NASA POWER (MERRA-2)	Central Delhi, daily, 2018–2025	2,922 days
Festival and policy dates	Public calendars, GRAP orders	2018–2025	8 Diwalis, 8 Dussehras, 2 GRAP seasons

The four stations span Delhi from east (Anand Vihar) to north (Ashok Vihar, Bawana) to south (Aya Nagar). A day counts for a station only when at least 18 of its 24 hours have valid readings. Under that rule, 10,571 of 11,688 possible station-days (90%) are usable.

A problem in the government data

Early in the work, two stations that should have shown different air turned out to have identical readings hour after hour. The CPCB download portal was serving Anand Vihar's data under other station names, including Ashok Vihar. Had this gone unnoticed, the study would have counted one station several times and reported falsely precise results.

The Ashok Vihar files were downloaded again and checked. The pipeline now compares every pair of stations hour by hour, and it stops if any two are identical in more than half of their shared hours. In the final data, the largest overlap between any two stations is 1.4%. The stations agree as neighbouring monitors should (hourly correlations of 0.76 to 0.90) without being copies of each other.

Fire data

Fire detections were kept only if they were:

- classified as vegetation fires;
- inside a box covering Punjab and Haryana (27.6–32.6°N, 73.8–77.6°E);
- more than 40 km from central Delhi, so that urban industrial heat sources are excluded.

In October–November, the S-NPP satellite recorded between 10,033 (2025) and 96,664 (2021) such detections a year.

4. How the study separates smoke from everything else

The hard part of this question is that everything bad happens at once. Burning peaks in late October and November, which is also when the air turns still and cold, when Diwali often falls, and when traffic and heating rise. A simple comparison of smoky and clean months would blame the fires for all of it.

The idea: let the wind decide

The study relies on something the fires cannot control: which way the wind blows on a given day. On a day when the wind comes from the north-west, smoke from Punjab and Haryana travels towards Delhi. On a day when it comes from the south-east, the same fires burn but their smoke drifts away.

So every fire detection is given a weight between 0 and 1, depending on how directly that day's wind blows from it towards Delhi:

- a fire directly upwind counts fully (1);
- a fire at right angles to the wind counts 0;
- a fire directly downwind is counted separately, in a "downwind" index.

The downwind index is the key safeguard. If the smoke really drives Delhi's pollution, upwind fires should raise PM2.5 and downwind fires should not. If both appear to raise it, something else, such as the season itself, is doing the work. This kind of check is called a **placebo test**.

Comparing days within the same week

The season builds over weeks, so the model includes a separate baseline for every calendar week of every year. Each day is therefore compared only with other days in the same week. The slow rise of the season, the harvest calendar and any policy stage in force all cancel out. What remains is the day-to-day variation in wind direction.

The model also accounts for:

- temperature, humidity, rain and wind speed;
- the general wind direction;
- the day of the week;
- differences between the four stations;
- Diwali and Dussehra days.

It looks at fires on the same day and on the two previous days, because smoke takes time to arrive and to clear.

Uncertainty

The four stations share the same air on any given day, and pollution carries over from one day to the next. Treating every station-day as independent would make the results look more certain than they are. The study therefore uses **Driscoll–Kraay standard errors**, which allow for both kinds of dependence, and all confidence intervals reflect that.

Turning effects into a share

Once the effect of upwind fires is estimated, the smoke's contribution on each day is the effect multiplied by that day's upwind fire index. Averaged over the season and divided by average PM2.5, this gives the fire share. Its confidence interval accounts for the uncertainty in every estimated effect.

One change made along the way

The original plan estimated a single effect per fire detection across all eight years. When the full archive arrived, it became clear that this assumption did not hold. Detections fell about 90% after 2021, but the smoke did not (Section 5.2). The final model therefore lets the effect per detection differ between 2018–2023 and 2024–2025. This change was made after seeing the data. Every version, including the original single-effect model, is reported in Section 7 so that readers can judge the choice for themselves.

5. Findings

5.1 Crop fires: about 15% of the season

Upwind fires raise Delhi's PM2.5 in both periods. Downwind fires, the placebo, do not.

Period	Upwind fires: 3-day effect per 1,000 weighted detections	Downwind fires (placebo)
2018–2023	+31.6 μg/m³ (95% CI 11.4 to 51.7), p = 0.002	−0.9 (−27.3 to 25.5), p = 0.95
2024–2025	+241.3 μg/m³ (74.4 to 408.2), p = 0.005	−16.6 (−178.6 to 145.5), p = 0.84

Most of the effect comes on the day the wind brings the smoke (+24.5 in 2018–2023, $p < 0.001$). Any carry-over to the next day is smaller and less certain, and by the third day nothing measurable remains.

Translated into a share of what Delhi breathes:

Period	Average PM2.5 (μg/m³)	From crop fires (μg/m³)	Share (95% CI)
October–November, 2018–2025	179.7	26.4	**14.7%** (7.4–22.0%)
October only	123.0	16.1	13.1% (7.1–19.0%)
November only	237.6	37.0	15.6% (7.5–23.7%)
2018–2023	174.7	25.3	14.5% (5.2–23.8%)
2024–2025	194.6	29.7	15.3% (4.7–25.8%)

The average hides large swings from day to day. On a typical day, smoke is about 8.5% of PM2.5. On the smokiest tenth of days it is 34% or more, and on 86 of the 485 season days it was above a quarter. That range is consistent with the published daily figures in Section 2, which reach 40–58% on individual days. Year by year, the share ranged from 7.5% (2023) to 23.8% (2021).

The result is also close to the independent season-level estimates: 13–17% from Down To Earth and about 28 μg/m³ a day from the Centre for Science and Environment. Three methods arriving at similar numbers is reassuring, though none of them is exact.

5.2 Fewer detections, the same smoke

The most unexpected result came from comparing years.

Year	Satellite detections, Oct–Nov	Night-time detections	Median fire power (MW)	Fire share of PM2.5
2018	73,045	0.5%	4.8	12.3%
2019	54,334	0.4%	4.9	11.2%
2020	86,601	0.6%	5.0	20.4%
2021	96,664	0.3%	5.1	23.8%
2022	55,254	0.6%	4.8	11.7%
2023	37,572	0.9%	4.5	7.5%

Year	Satellite detections, Oct–Nov	Night-time detections	Median fire power (MW)	Fire share of PM2.5
2024	16,074	4.0%	3.8	18.2%
2025	10,033	3.7%	3.8	12.0%

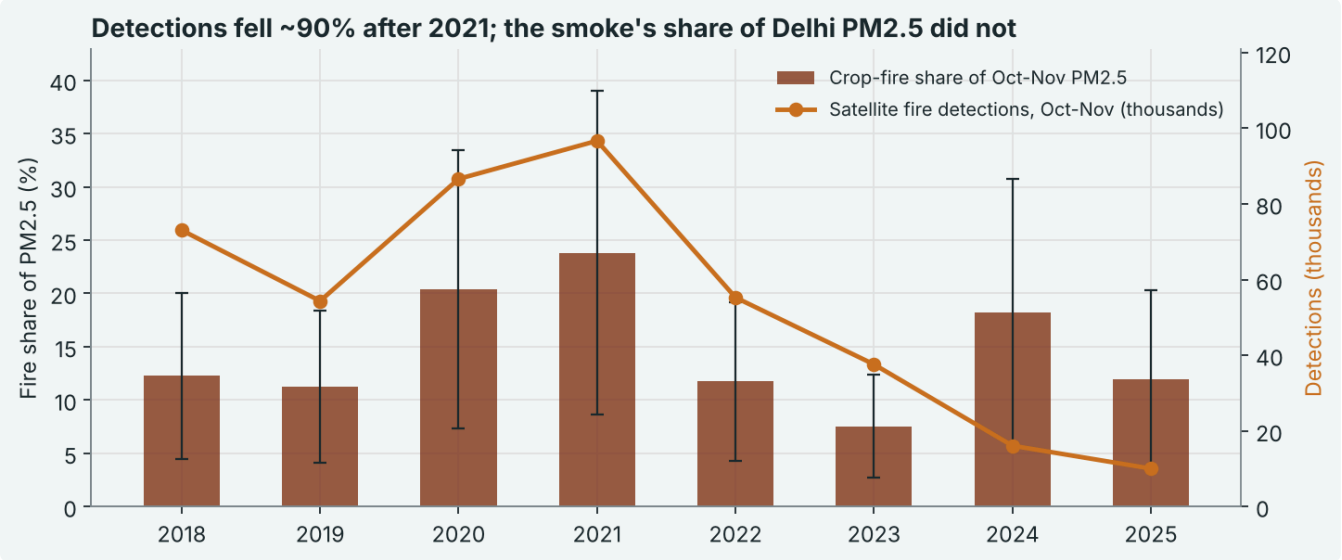


Figure 1. Bars: crop-fire share of October–November PM2.5 with 95% confidence intervals. Line: satellite fire detections. Detections fell sharply after 2021; the smoke's share did not.

Detections fell by about 90% between 2021 and 2025. Delhi's PM2.5 did not fall, and neither did the smoke's share. In 2024–2025, each detection was associated with about eight times more PM2.5 than in earlier years (241 vs 32 per 1,000; difference p = 0.014). At the same time the detections themselves changed: night-time detections rose from under 1% to about 4%, and typical fire power dropped.

The most likely explanation is that a growing share of burning now happens when the satellite is not looking. This matches the ISRO finding that peak burning moved from early afternoon to about 17:00. This study cannot prove that on its own, but the practical conclusion holds either way: **a fall in satellite fire counts is not, by itself, evidence that less smoke is reaching Delhi.** It is also why a single-effect model, which assumes every detection means the same thing in every year, would put the 2024–2025 share at only 2%.

5.3 Diwali and Dussehra

Diwali produces the sharpest spike of the season, but only for a few days.

Window	Diwali	Dussehra
Festival night, 20:00–02:00, compared with ordinary nights	500 vs 222 µg/m³ (2.3×), higher in all 8 years	140 vs 118 µg/m³ (1.2×), higher in 6 of 8 years

Window	Diwali	Dussehra
Worst single hour	up to 7.0× an ordinary hour (2025)	up to 1.8×
Festival day and the next day	about 26% of PM2.5	about 1%
The week starting on the festival	about 10% (95% CI 1–20%)	about 1%
The month it falls in	about 3.4% (0.3–6.6%)	about 0.6%
The October–November season	0.5–1.7%, depending on method	about 0%

The spike fades within about a day and a half. The extra PM2.5 is +190 µg/m³ in the evening, +260 after midnight, +97 by the next morning and +26 by the afternoon.

The wind makes a large difference to Diwali. In the five years with light wind on the festival (2018–2021 and 2025), Diwali accounted for about 17% of the following week. In the three breezier years (2022–2024), its effect was close to zero. This agrees with the IIT Delhi finding that fireworks dominate the air's chemistry during the festival, while biomass smoke takes over in the days after.

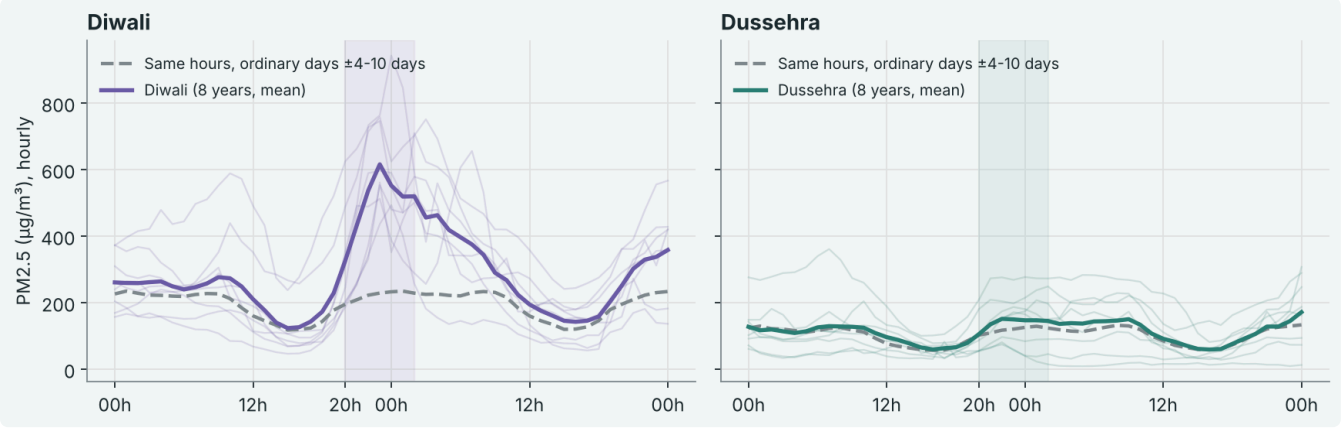


Figure 2. Hourly PM2.5 around each festival (thin lines: individual years; thick line: average of 8 years) compared with the same hours on ordinary days 4–10 days away (dashed). The shaded band marks 20:00–02:00.

Dussehra, despite the effigy burning, left no detectable mark on the four monitors.

5.4 Weather

Weather does not create pollution, but it decides how much of it stays over the city. Within the same week, a still, humid, stagnant day carries about **53 µg/m³ more PM2.5** than a well-ventilated one (comparing the 10th and 90th percentiles of the combined

weather effect). That swing is about 30% of the season's average. No single weather variable is decisive on its own; it is the combination that matters.

5.5 Putting the season together

Source	Share of October–November PM2.5	How it was obtained
Crop fires in Punjab and Haryana	about 15% (7–22%)	Measured in this study
Diwali and Dussehra	under 1% (regression) to under 2%	Measured in this study
Delhi's own emissions	about 36% (range 26–50%)	Taken from TERI–ARAI (2018), not measured here
The rest: NCR, regional sources and particles formed in the air	about half	What remains

The table should be read with care. Only the first two rows were measured in this study. Delhi's own share comes from an outside source, and the last row is simply what is left over.

6. Would acting a month earlier help?

It is tempting to think that starting the emergency plan sooner would prevent the worst weeks. The numbers suggest it would help only a little.

6.1 Starting GRAP before the burning season

Burning begins in October, so "a month before" means GRAP Stage III in force from 1 September instead of mid-November. This study cannot measure GRAP's effect itself, so it borrows the full range from the earlier GRAP evaluation and carries that uncertainty through:

Season	GRAP actually began	If it began on 1 September	Best case	At the earlier study's central estimate
2024–25	15 Nov 2024	75 days earlier	3.4% lower PM2.5 over Sep–Jan	no detectable change

Season	GRAP actually began	If it began on 1 September	Best case	At the earlier study's central estimate
2025–26	12 Nov 2025	72 days earlier	3.7% lower	no detectable change

Starting on 1 October instead gives a best case of 2.9–3.1%. Even the best case is small, because the plan's own measured effect is small and uncertain. The central reading of the earlier study is no detectable effect. Its positive point estimate most likely reflects that GRAP is switched on when the air is already at its worst, not that GRAP makes the air dirtier.

6.2 Real cuts in October

Cutting emissions does more than moving a calendar. Two kinds of cut were examined:

Action from 1 October	Expected fall in October PM2.5	Basis
Delhi's own emissions cut by 10%	about 3.6% (2.6–5.0%)	Scenario using TERI–ARAI's 26–50% local share
Delhi's own emissions cut by 30%	about 10.8% (7.8–15.0%)	as above
Delhi's own emissions cut by 50%	about 18% (13–25%)	as above
October crop fires halved	about 6.5% (3.6–9.5%)	Measured October fire share in this study
October crop fires stopped entirely	about 13% (7–19%)	as above

The local-emission rows are scenarios, not measurements. They show what a given cut would mean if the outside estimate of Delhi's own share is right.

6.3 When the worst days actually happen

Of the 106 severe days (above 250 μg/m³) in October–November 2018–2025, only 10 came before 1 November. Most of the emergency-level pollution arrives in November, when fires, still air and colder nights coincide.

October is not clean, though. Its average ranged from 81 to 149 μg/m³ across these years, 1.3 to 2.5 times the daily standard, every year. Earlier action would mainly improve the long stretch of unhealthy but not "severe" days in October, rather than prevent November's emergencies.

7. How sure can we be?

A single estimate is easy to produce. The harder question is whether it survives attempts to break it. The study ran the main result through a series of checks, and reports all of them, including the ones that did not go cleanly.

7.1 Tests that support the result

Test	What it asks	Result
Downwind placebo	Do fires blowing away from Delhi also seem to raise PM2.5?	No: placebo share −0.2% (CI −2.1 to 1.7%)
Permutation test	If wind directions are shuffled at random within each week, 1,000 times, how often is a share this large found by chance?	2% of the time (p = 0.020)
Total fires held fixed	With the number of fires fixed, does wind alignment alone matter?	Yes: 14.6% (1.7–27.4%)
Event study	What happens to PM2.5 around the heaviest upwind-fire days, compared with the heaviest downwind-fire days?	+44 μg/m³ on the day (CI 21–68) vs −13 for downwind days
Dose–response	Do heavier smoke days add more?	Yes, but mostly on the heaviest fifth of days (+38 μg/m³, p = 0.028)
Station by station	Does it hold at each monitor?	Yes, 10–21% at every station, each above zero
Second satellite	Does a different satellite (NOAA-20, 2024–2025) agree?	Broadly: 13.1% vs 17.8% with the main satellite

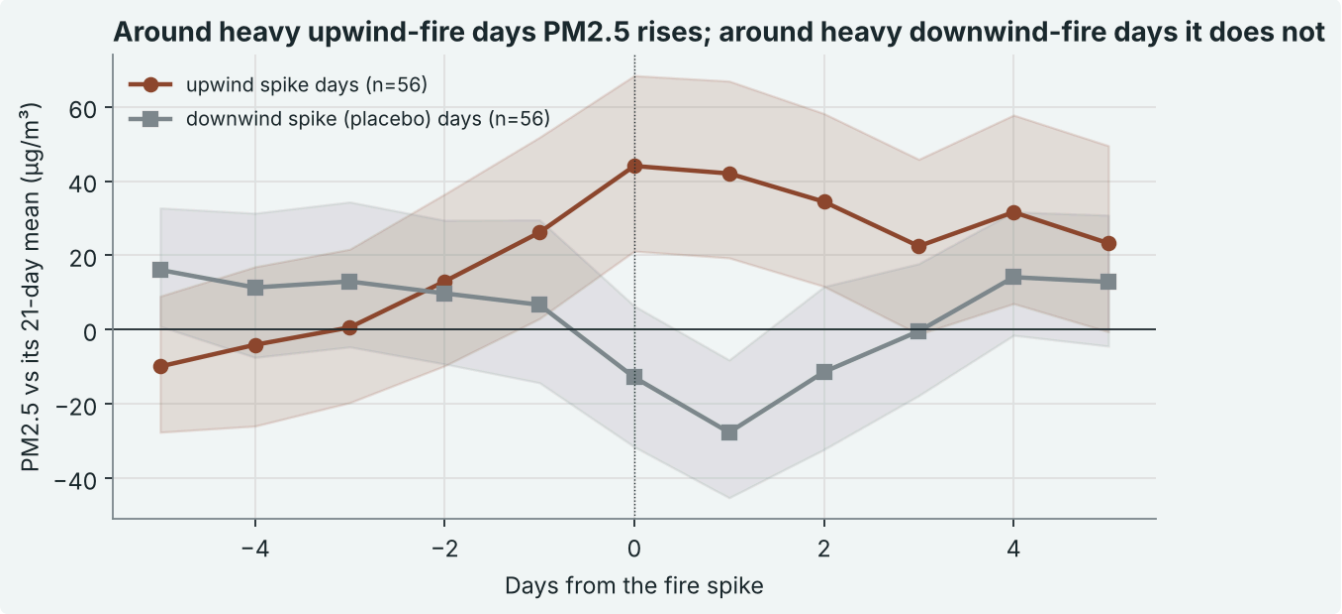


Figure 3. PM2.5 relative to its 21-day average, from five days before to five days after the heaviest 10% of upwind-fire days (brown) and downwind-fire days (grey), with 95% bootstrap intervals.

The share stays between 12% and 18% across other reasonable choices:

- 10 m instead of 50 m wind;
- a narrower wind cone;
- high-confidence fires only;
- a wider season window (15 September to 15 December);
- three or eight detection periods instead of two;
- an extra day of lag.

The original single-effect model gives 10–12% overall, but only about 2% for 2024–2025, which is the undercount described in Section 5.2.

7.2 Tests that did not go cleanly

Future fires in 2024–2025. Fires detected one or two days ahead should not affect today's air. In 2018–2023 they do not ($p = 0.22$). In 2024–2025 they do ($p < 0.001$). The most plausible reason is the same timing shift: burning that today's satellite pass misses gets detected tomorrow, so tomorrow's count partly measures today's fires. Adding future fires to the model leaves the share unchanged at 14.8%. Still, this is a warning that recent fire counts are a noisier measure of burning than older ones.

Carbon monoxide. Smoke contains carbon monoxide, so CO was expected to rise with upwind fires as a confirming signal. It did not (+0.06 mg/m³, $p = 0.11$). In Delhi, CO is dominated by traffic and local combustion, which probably drowns out the smaller contribution from distant smoke.

The monitors' own wind sensors. Repeating the analysis with wind measured at the four stations instead of NASA's regional wind gives almost no effect. On closer inspection, those rooftop sensors disagree with the regional wind direction by a median of 95°, and only 20% of days fall within 45°. Sheltered sensors in a dense city cannot tell which fields are upwind, so this check is uninformative rather than contradictory.

The permutation baseline. With shuffled winds, the share averages 4% rather than zero. This is because shuffled winds still partly follow the weekly fire total. The real result (14.6%) is well above that baseline, but the gap is smaller than the raw number suggests.

8. What this study cannot tell us

It is as important to be clear about the limits as about the findings.

- **Four monitors are not the whole city.** The stations cover east, north and south Delhi, and the results agree across them, but other neighbourhoods may differ.
- **Delhi's own share was not measured.** The 36% figure for local emissions comes from TERI–ARAI (2018) and may have changed since. The local-cut scenarios in Section 6 inherit that uncertainty.
- **The per-period model was a choice made after seeing the data.** It is well motivated by the change in detections and the ISRO timing study, and all alternatives are shown. But it was not planned in advance.
- **Satellite fire counts are an imperfect measure of burning,** especially in recent years. The study measures the link between detected fires and Delhi's air, not burning directly.
- **Linearity.** The model assumes each additional detection adds the same amount within a period. Shares on single extreme days are therefore rough; averages over the season are the reliable figures.
- **The GRAP counterfactual is borrowed.** It applies the earlier study's estimate rather than measuring GRAP afresh, and that estimate was itself uncertain.
- **The Diwali shares depend on method.** Comparing with nearby ordinary days gives a larger seasonal share (1.7%) than the weather-controlled regression (0.5%). Both are small, but neither is exact.

9. What it means

The picture that emerges is less dramatic than either side of the yearly argument, and more useful.

Crop fires are a genuine and measurable cause of Delhi's autumn pollution. On the worst days they account for a third or more of what people breathe. Over the season they account for about one part in seven. Removing every fire would not give Delhi clean air. It would still leave October and November far above the national standard.

Diwali deserves its reputation for one terrible night. In every year studied, the air after the fireworks was more than twice as bad as an ordinary night. But because the spike passes within a day or two, it is not what makes the season as a whole so polluted.

The weather is the multiplier. It does not create the smoke or the dust, but on still days it traps them. That is part of why the same emissions feel so much worse in November than in October.

Several practical points follow, each stated only as strongly as the evidence supports:

1. **Start real cuts in October.** Moving the emergency plan earlier offers little. Sustained reductions in Delhi's own emissions, starting from 1 October, would lower October's PM2.5 roughly in proportion to the cut.
2. **Keep working on crop burning, and measure it better.** Halving October fires would lower October's PM2.5 by about 6.5%. Because daytime satellite counts now miss more burning than before, they should not be the main yardstick of progress. Night-time and geostationary observations, and checks on the ground, would give a truer picture.
3. **Use wind forecasts.** Smoke arrives on the same day the wind turns north-westerly. Days with active fires and forecast north-west winds are the days to warn people and to tighten local restrictions in advance.
4. **Treat Diwali as a short, predictable spike.** Protecting people on the night and the day after is reasonable. It will not, by itself, change the season.

Conclusion

Over eight seasons, crop-fire smoke accounts for about 15% of Delhi's October–November PM2.5. The evidence that it is causal is strong: upwind fires raise pollution, downwind fires do not, and shuffled winds rarely reproduce the effect. That share held steady even as satellite fire counts fell by 90%, so falling counts should not be read as falling smoke. Diwali causes the season's sharpest spike, but it fades within days and adds little to the season overall. Starting the emergency plan a month earlier would change little; steady emission cuts from October, in the city and in the fields, are what the evidence points to.

Appendix

A. The model

For station s on day d, with detection periods k (2018–2023 and 2024–2025):

$$\text{PM}_{s,d} = \sum_{k}\sum_{l=0}^{2}\beta_{lk}\,\text{FireUp}_{d-l}\,\mathbb{1}[k] + \sum_{k}\sum_{l=0}^{2}\gamma_{lk}\,\text{FireDown}_{d-l}\,\mathbb{1}[k] + \mathbf{W}_d'\delta + \text{Fest}_d'\phi + \alpha_s + \lambda_{\text{week}\times\text{year}} + \omega_{\text{weekday}} + \varepsilon_{s,d}$$

$$\text{FireUp}_d = \frac{1}{1000}\sum_{i} \max\big(0,\ \cos(\theta_d - b_i)\big), \qquad \text{FireDown}_d = \frac{1}{1000}\sum_{i} \max\big(0,\ -\cos(\theta_d - b_i)\big)$$

θ is the direction the 50 m wind blows from, and b is the bearing from Delhi to fire i. W holds the weather controls and wind sector, and Fest holds the Diwali and Dussehra day dummies. The model is fitted on 1,822 station-days (485 days) by ordinary least squares, with Driscoll–Kraay standard errors (14-day Bartlett kernel). The fire share is the season average of the summed products β × FireUp, divided by average PM2.5, with a delta-method confidence interval.

B. Statistical tests

Test	Statistic	Result	p-value
Upwind effect, 2018–2023	Wald z-test	+31.6 per 1,000 detections	0.002
Upwind effect, 2024–2025	Wald z-test	+241.3	0.005
Downwind placebo, both periods	Wald z-test	−0.9; −16.6	0.95; 0.84
Difference between periods	Wald z-test	+209.7	0.014
Permutation (1,000 runs)	Share under shuffled wind	Real 14.6% vs shuffled mean 4.1%	0.020
Future-fire placebo	Wald z-test	2018–23: +10.7; 2024–25: +105.7	0.22; < 0.001
Heaviest dose fifth	Wald z-test	+37.9 μg/m³	0.028

Test	Statistic	Result	p-value
Diwali night above baseline	Exact binomial sign test	8 of 8 years	0.008
Diwali, 2-day regression	Wald z-test	+53.7 µg/m³	0.22
Dussehra, 2-day regression	Wald z-test	−6.8 µg/m³	0.66
CO tracer	Wald z-test	+0.06 mg/m³	0.11

C. Reproducibility

The full analysis is written in Python (numpy, pandas, matplotlib). It runs end to end with one command, `python run_all.py`, in under a minute including 1,000 permutations. All festival tests are in a separate module, `src/diwali.py`. Nine unit tests check the fire geometry, lags, period terms, the sign test and the GRAP calculation. Every number in this report can be traced to a table in `reports/tables/`.

Code and data: github.com/Aishwaryasrivastav31/delhi-winter-smog

D. References

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